# THREE-WAY EQUIVALENCE FOR FLAT MAPS WITH REGULAR FIBRE

THE STACKS PROJECT AUTHORS

*Editorial note.* 'The implication (1) => (2) is trivial' is definitional (tor amplitude bounds the Tor vanishing range); retained verbatim.

*Allowed standard prerequisites (editorial).*

- Regular sequence splitting. lemma-regular-sequence (same chapter, named). Drives the (2) => (3) direction after the tor-dimension input.
- Perfect over regular local. more-algebra-lemma-perfect-over-regular-local-ring (cross-chapter, named). Converts finitely many nonzero Tor against the residue field into finite tor dimension over S/IS.
- Koszul resolution and tor amplitude transfer. more-algebra-lemma-flat-push-tor-amplitude (cross-chapter, named). Supports the Koszul-resolution argument in (3) => (1).

*Source reference.* Original reference 00LM: focused-seeds/geometry/S9-dpa-perfect-map-ci/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original lemma 09PW: sources/stacks-dpa/pinned-original-source.tex, source lines 1811–1811. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 066J: focused-seeds/geometry/S9-dpa-perfect-map-ci/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 066A: focused-seeds/geometry/S9-dpa-perfect-map-ci/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 0669: focused-seeds/geometry/S9-dpa-perfect-map-ci/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof

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has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 09CC: focused-seeds/geometry/S9-dpa-perfect-map-ci/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

*Source reference.* Original reference 09PC: focused-seeds/geometry/S9-dpa-perfect-map-ci/ref-tags.txt, source lines 10–10. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

**Lemma 0.1.** *Let $R \to S$ be a local ring map of Noetherian local rings. Let $I \subset R$ and $J \subset S$ be ideals with $IS \subset J$. If $R \to S$ is flat and $S/\mathfrak{m}_R S$ is regular, then the following are equivalent*

(1) *$J$ is generated by a regular sequence and $S/J$ has finite tor dimension as a module over $R/I$,*
(2) *$J$ is generated by a regular sequence and $Tor_p^{R/I}(S/J, R/\mathfrak{m}_R)$ is nonzero for only finitely many $p$,*
(3) *$I$ is generated by a regular sequence and $J/IS$ is generated by a regular sequence in $S/IS$.*

*Proof.* If (3) holds, then $J$ is generated by a regular sequence, see for example More on Algebra, Lemmas 0669 and 09CC. Moreover, if (3) holds, then $S/J = (S/I)/(J/I)$ has finite projective dimension over $S/IS$ because the Koszul complex will be a finite free resolution of $S/J$ over $S/IS$. Since $R/I \to S/IS$ is flat, it then follows that $S/J$ has finite tor dimension over $R/I$ by More on Algebra, Lemma 066J. Thus (3) implies (1).

The implication (1) $\Rightarrow$ (2) is trivial. Assume (2). By More on Algebra, Lemma 09PC we find that $S/J$ has finite tor dimension over $S/IS$. Thus we can apply Lemma 09PW to conclude that $IS$ and $J/IS$ are generated by regular sequences. Let $f_1, \ldots, f_r \in I$ be a minimal system of generators of $I$. Since $R \to S$ is flat, we see that $f_1, \ldots, f_r$ form a minimal system of generators for $IS$ in $S$. Thus $f_1, \ldots, f_r \in R$ is a sequence of elements whose images in $S$ form a regular sequence by More on Algebra, Lemmas 066A and 09CC. Thus $f_1, \ldots, f_r$ is a regular sequence in $R$ by Algebra, Lemma 00LM. $\square$